

## **[BUG] ADAPT ansatz not removing superfluous qubits**

<https://github.com/goodchemistryco/Tangelo/issues/186>

ROW 73

# [BUG] ADAPT ansatz not removing superfluous qubits #186

New issue

Closed



KrzysztofB-1qbit opened on Aug 8, 2022

...

## Issue: Bug Report

### Expected Behavior

When generating an ADAPT ansatz with scbk, there should be 2 qubits removed

### Current Behavior

ADAPT returns a circuit with superfluous qubits which are not actually used for anything. These can be removed using the `trim_qubits` method of the `Circuit` class.

### Steps to Reproduce (minimal example)

The ansatz in the project Neon should be a 2 qubit circuit, but it clearly indicates a width of 4 qubits, even though there are only gates on the first 2.

### Environment

MacOS 11.6.1



KrzysztofB-1qbit added **bug** on Aug 8, 2022



KrzysztofB-1qbit assigned [ValentinS4t1qbit](#), [alexfleury-sb](#) and [JamesB-1qbit](#) on Aug 8, 2022



ValentinS4t1qbit on Aug 8, 2022

Collaborator

Thank you for reporting this !! I think you raise an important point we may not have looked into: meta VQEs or complex algorithms in general may not have a constant number of qubits. Maybe it can grow, maybe it can go down as well. When going to a different step or concatenating circuits, maybe we should try trimming once in a while and updating qubit number if relevant to ensure we keep resources low during runtime.

I think it's probably a small surgery we can do inside the code of ADAPT itself, unless we foresee VQE itself may have fancy ansatzes whose number of qubits-in-use may change because... stuff.



alexfleury-sb mentioned this on Sep 15, 2022

[Superfluous qubits with scBK in ADAPTSolver #214](#)



alexfleury-sb on Sep 16, 2022

Collaborator

Closed with [#214](#).



alexfleury-sb closed this as **completed** on Sep 16, 2022

### Assignees

[JamesB-1qbit](#)

[ValentinS4t1qbit](#)

[alexfleury-sb](#)

### Labels

**bug**

### Type

No type

### Projects

No projects

### Milestone

No milestone

### Relationships

None yet

### Development

No branches or pull requests

### Notifications

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### Participants



# Superfluous qubits with scBK in ADAPTSolver #214

<> Code

**Merged** ValentinS4t1qbit merged 1 commit into `sandbox-quantum:develop` from `alexfleury-sb:issue_186` on Sep 15, 2022

Conversation 1 Commits 1 Checks 0 Files changed 2 +3 -4



alexfleury-sb commented on Sep 15, 2022

Collaborator

Superfluous qubits (2) were initialized in ADAPTSolver with the scBK mapping. This fix issue #186.



Fixed number of qubits with scBK in ADAPT.

05d975e

alexfleury-sb requested review from ValentinS4t1qbit and JamesB-1qbit as code owners 3 years ago



JamesB-1qbit approved these changes on Sep 15, 2022

View reviewed changes



ValentinS4t1qbit approved these changes on Sep 15, 2022

View reviewed changes

ValentinS4t1qbit left a comment

Collaborator

Looks pretty simple.  
You have not added a test for the particular scenario that motivated it, but im not sure that's necessary.



ValentinS4t1qbit merged commit c1d1df4 into `sandbox-quantum:develop` on Sep 15, 2022

alexfleury-sb mentioned this pull request on Sep 16, 2022

[BUG] ADAPT ansatz not removing superfluous qubits #186

Closed

alexfleury-sb deleted the `issue_186` branch 3 years ago

ValentinS4t1qbit added a commit that referenced this pull request on Nov 9, 2022

New release v0.3.3 into main (#248)

Partially verified

e985661

## Reviewers

JamesB-1qbit



ValentinS4t1qbit



## Assignees

No one assigned

## Labels

None yet

## Projects

None yet

## Milestone

No milestone

## Development

Successfully merging this pull request may close these issues.

None yet

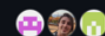
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